

Cyclistic Bike-Share Case Study

How Does a Bike-Share Navigate Speedy Success?

Prepared by: Aseel Alnoor

Role: Junior Data Analyst

Date: May 2026

The Business Task

Objective: Analyze historical trip data to identify behavioral differences between "Casual Riders" and "Annual Members."

Goal: Design data-driven marketing strategies to convert casual riders into annual members, ensuring long-term financial growth for Cyclistic.

Data Sources

Dataset: Cyclistic's historical trip data (last 12 months).

Type: Public data provided by Motivate International Inc.

Integrity: The data is clean, comprehensive, and anonymized to protect user privacy (no PII included).

Documentation of the cleaning

Tools: Python (Pandas library) was used for data cleaning and manipulation.

Steps: Filtered out trips shorter than 1 minute or longer than 24 hours.

Checked for and removed duplicate entries and null values.

Formatted columns (Dates, Station names, and User types) for consistency.

Calculated "Ride Length" and "Day of the Week" for deeper analysis.

Summary of Analysis

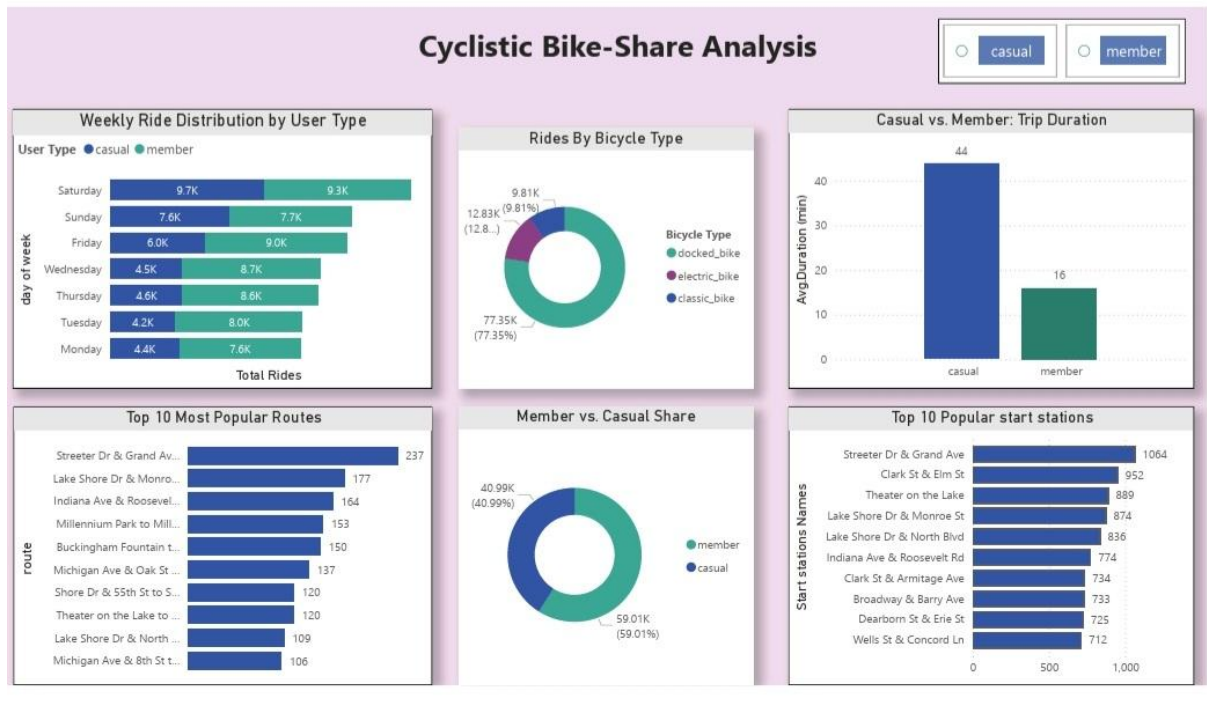
User Behavior: Casual riders take significantly longer trips on average compared to members.

Temporal Patterns: Casual rider activity peaks during weekends (Saturday & Sunday), while members show steady usage throughout the workweek.

Top Stations: Identified the most popular start/end stations for casual riders to target marketing efforts.

Visualization

Key Insight: The dashboard clearly illustrates the surge in casual riders during the weekend and their preference for certain types of bikes and specific tourist-heavy stations.



Top Three Recommendations

Weekend Membership Plan: Launch a "Weekend-Only" annual membership tier to attract casual riders who primarily use the service for leisure on Saturdays and Sundays.

Long-Duration Rewards: Since casual riders take longer trips, introduce membership benefits that offer discounts or "loyalty points" for rides exceeding 30 minutes.

Targeted Seasonal Campaigns: Increase digital marketing and physical advertisements at the "Top 10 Stations" during peak summer weekends, highlighting the cost savings of moving from a single-ride to an annual plan.